MILITARY SPECIFICATION

GYROMAGNETIC COMPASS SET AN/ASN-43()

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 This specification covers one type of Gyromagnetic Compass Set, designated AN/ASN-43(), to be used as a universal heading reference system in fixed and rotary wing aircraft using gyromagnetic compass systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
MIL-T-45277	Transmitter, Induction Compass T-611()/ASN
MIL-G-45279	Gyro, Directional CN-998()/ASN-43
MIL-G-45281	Gyromagnetic Compass Set AN/ASN-43(), Packaging and Packing of
MIL-C-55398	Compensator, Magnetic Flux CN-405()/ASN
MIL-M-13231	Marking of Electronic Items

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-252	Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired, and Other Devices
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the Contracting Officer.)

FSC 6605

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3. REQUIREMENTS

3.1 Description. The Gyromagnetic Compass Set AN/ASN-43() consists of a Directional Gyro CN-998()/ASN-43; Transmitter, Induction Compass, Type T-611()/ASN; and Compensator, Magnetic Flux, Type CN-405()/ASN. (Included in the functional tests of this specification, but not a part of the compass set, is an Indicator, Radio-Magnetic Compass ID-998/ASN, and Amplifier, Electronic Control AM-3209/ASN.) The compass set provides accurate stabilized magnetic heading through 360 degrees in azimuth. Gyroscope stabilized reference information for automatic pilot or other flight equipment is avaiable from a synchro transmitter in the directional gyroscope. The operation of the system(See 6.3) shall be such that it may function in a magnetically slaved(MAG) mode or in a latitude-corrected free gyro(DG) mode.

3.1.1 Directional Gyro CN-998()/ASN-43 shall be in accordance with MIL-G-45279(EL).

3.1.2 Transmitter, Induction Compass, Type T-611()/ASN shall be in accordance with MIL-T-45277(EL).

3.1.3 Compensator, Magnetic Flux, Type CN-405()/ASN shall be in accordance with MIL-C-55398(EL).

3.2 Marking.

3.2.1 General. Marking shall conform to MIL-M-13231 and the equipment drawings (See 4.9).

3.2.2 Visibility. Whenever practicable, parts shall be so mounted that identification markings will be readily visible with minimum disassembly of the equipment.

3.3 Construction. The equipment shall be constructed in accordance with the requirements of this document and the requirements and drawings specified in MIL-C-55398(EL), MIL-T-45277(EL), and MIL-G-45279(EL).

3.4 Interchangeability. Like items between sets shall be physically and functionally interchangeable without modification of the items. Like items between sets shall conform to Requirement 7 of MIL-STD-454(See 4.8).

3.5 Operational. The Gyromagnetic Compass Set shall be operated as specified after applicable processes and inspections are completed to determine that each unit is operable prior to packaging(See 4.7).

3.5.1 Primary Input Power Requirements. The Gyromagnetic Compass Set shall meet all applicable requirements and shall give specified

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performance when operated from an AC power source with characteristics and limits as defined in MIL-STD-704. The AC power requirement shall be 115 volts, 400 Hz, single phase, Category "B", 0.5 amperes maximum (See 4.7.1).

3.5.2 Slaving Rate. The gyroscope shall precess at a rate of 2.5 ± 1.0 degrees per minute when the magnetic heading is changed. (See 4.7.2.1)

3.5.3 Static Settling. The heading indication shall be within ± 1.0 degree for a ten degree clockwise and counterclockwise change in magnetic heading position, after synchronized condition is attained. The output angular readings shall not differ by more than 0.25 degrees from each other. The two heading indications shall settle to within 1.0 degree of the synchronized heading recorded above (after removal of induction compass transmitter index and angle position indicator errors) and shall not differ by more than 0.25 degree(See4.7.2.2).

3.5.4 Scale Error. The scale error of the compass set shall not exceed 0.75 degrees rms for all headings(See 4.7.2.3).

3.6 Workmanship. The equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454, MIL-C-55398(EL), MIL-T-45277(EL), and MIL-G-45279(EL)(See 4.9).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of Inspection. Inspection shall be classified as follows:

Quality Conformance Inspection.

(1) Quality Conformance Inspection of equipment before preparation for delivery(See 4.5).

(2) Quality Conformance Inspection of preparation for delivery(See 4.10).

4.3 First Article Inspection. Not required.

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4.4 Inspection Covered by Subsidiary Documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract:

ITEM	WHERE REQUIRED
Directional Gyro	3.1.1
Transmitter	3.1.2
Compensator	3.1.3
Marking	3.2

4.5 Quality Conformance Inspection of Equipment Before Preparation for Delivery. The contractor shall perform the inspections specified in 4.4, 4.5.1, and 4.5.2. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A Inspection. Each gyromagnetic compass set on contract shall be inspected to the inspections specified in Table I. Discrete lots shall be formed from sets that pass this inspection. Factors of lot composition not defined herein, or in the contracts, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels and AQL's indicated in Table I.

TABLE I - GROUP A INSPECTION

INSPECTION	REQT PARA	TEST PARA	AQL MAJOR	AQL MINOR
Visual and Mechanical	3.6	4.9	1%	4%
Operational	3.5	4.7	1%	--

4.5.2 Group B Inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table II and the special inspection level S4 of MIL-STD-105.

TABLE II - GROUP B INSPECTION

INSPECTION	REQT PARA	TEST PARA	AQL
Interchangeability	3.4	4.8	6.5%

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4.5.3 Group C Inspection. Not required.

4.6 Test Conditions.

4.6.1 Conditions. Unless otherwise specified, all inspections required by this specification shall be made under the following conditions:

a. Temperature - Ambient.

b. Pressure - Ambient.

c. Humidity - Ambient.

d. Input Power - 115 ± 2 volts, AC, 400 ± 10 Hz, single phase.

4.6.2 Attitude. Unless otherwise specified, the compass set shall be tested in its normal operating position.

4.7 Operational.

4.7.1 Primary Input Power (See3.5.1). Power shall be applied to the gyromagnetic compass system and the following tests performed at the following combinations of voltage and frequency:

a. 107.5 ± 1 volts at 380 ± 2 Hz.

b. 107.5 ± 1 volts at 420 ± 2 Hz.

c. 119.5 ± 1 volts at 380 ± 2 Hz.

d. 119.5 ± 1 volts at 420 ± 2 Hz.

e. 115 ± 1 volts at 380 ± 2 Hz.

f. 115 ± 1 volts at 420 ± 2 Hz.

4.7.2 Operational Test. An operational test, consisting of the following tests shall be performed on the AN/ASN-43 set, using the CN-998()/ASN-43 Directional Gyro, the CN-405()/ASN Magnetic Flux Compensator and the T-611()/ASN Induction Compass Transmitter. The test shall be made with a Radio Magnetic Compass Indicator ID-998/ASN and an Electronic Control Amplifier AM-3209/ASN, and with the units connected as shown in Figure 1 and 2. Unless othewise specified, the compass shall be operated in the slave mode. A 10 minute warm-up period shall be allowed.

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4.7.2.1 "Slaving Rate" (See 3.5.2). With the system synchronized to a North (0^0) heading, rotate the inductor compass transmitter to a 30 degree heading and measure the time required for the system to precess from 5 through 15 degrees. Resynchronize the system to a north (0^0) transmitter heading; rotate the transmitter to a 330 degree heading and measure the time required for the system to precess from 355 degrees through 345 degrees.

4.7.2.2 Static Settling(See 3.5.3). With the induction compass transmitter positioned to a north (0^0) heading, the sync knob on the ID-998/ASN Indicator shall be rotated in the direction indicated on the knob until the annunciator on the ID-998/ASN Indicator is centered. After recording the heading on the angle position indicator (low and high excursion reading taken and averaged), rotate the sync knob to produce a reading of 10 degrees on the angle position indicator. Repeat the above except the sync knob shall be rotated to produce a reading of 350 degrees on the angle position indicator.

4.7.2.3 Scale Error(See 3.5.4). The Gyro shall be level in a static position during this test. Record the error between the induction compass transmitter heading and the angle position indicator reading at 30 degree heading increments starting at 0 degree. The system shall be resynchronized on each heading by rotating the sync knob on the ID-998/ASN Indicator. The RMS error shall be computed as follows:

$$\text{RMS Error} = \sqrt{\frac{\sum (E)^2}{12}}$$

where E is the difference between the transmitter heading and angle position indicator reading on each of twelve 30 degree heading increments.

4.8 Inspection for Interchangeability. Each unit of each sample set shall be interchanged with a like unit and an operational test(4.7) shall be performed to determine compliance with 3.4.

4.9 Visual and Mechanical Inspection. The set shall be examined for the defects listed for Type 7, Finish and Type 8, Marking in MIL-STD-252.

4.10 Quality Conformance Inspection of Preparation for Delivery. Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY

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5.1 Preservation and Packaging. Preservation, packing, marking and testing shall be in accordance with MIL-G-45281.

6. NOTES

6.1 Intended Use. The compass set covered by this specification is intended for use in rotary and fixed wing aircraft requiring gyro-stabilized magnetic heading of nominal accuracy. The set is capable of supplying magnetic heading or gyro-stabilized reference information as required for automatic stabilization, navigation, and surveillance systems.

6.2 Ordering Data. Procurement documents shall specify the following:

a. Title, number, and date of this document and any amendment thereto.

b. Level of packaging and level of packing required for shipment(Level A, Level B, or Level C), and such additional marking that may be required.

c. The specific paragraphs of MIL-G-45281 which are applicable to the particular procurement.

d. First Article inspection not required for the Gyromagnetic Compass Set AN/ASN-43 (See 3.1.1, 3.1.2, and 3.1.3).

e. Place of final inspection.

f. Rough handling test(Preparation for Delivery, if required).

6.3 Definitions. All references to the words "system" or "compass system" in this document shall be defined as the Gyromagnetic Compass Set AN/ASN-43( ) when used with Indicator, Radio-Magnetic Compass ID-988/ASN and Amplifier, Electronic Control AM-3209/ASN(See 3.1).

6.4 Testing Order. Performance testing of the Gyromagnetic Compass Set AN/ASN-43( ) shall be performed after completion of the inspections and test on the individual components in accordance with their specifications (See 4.4).

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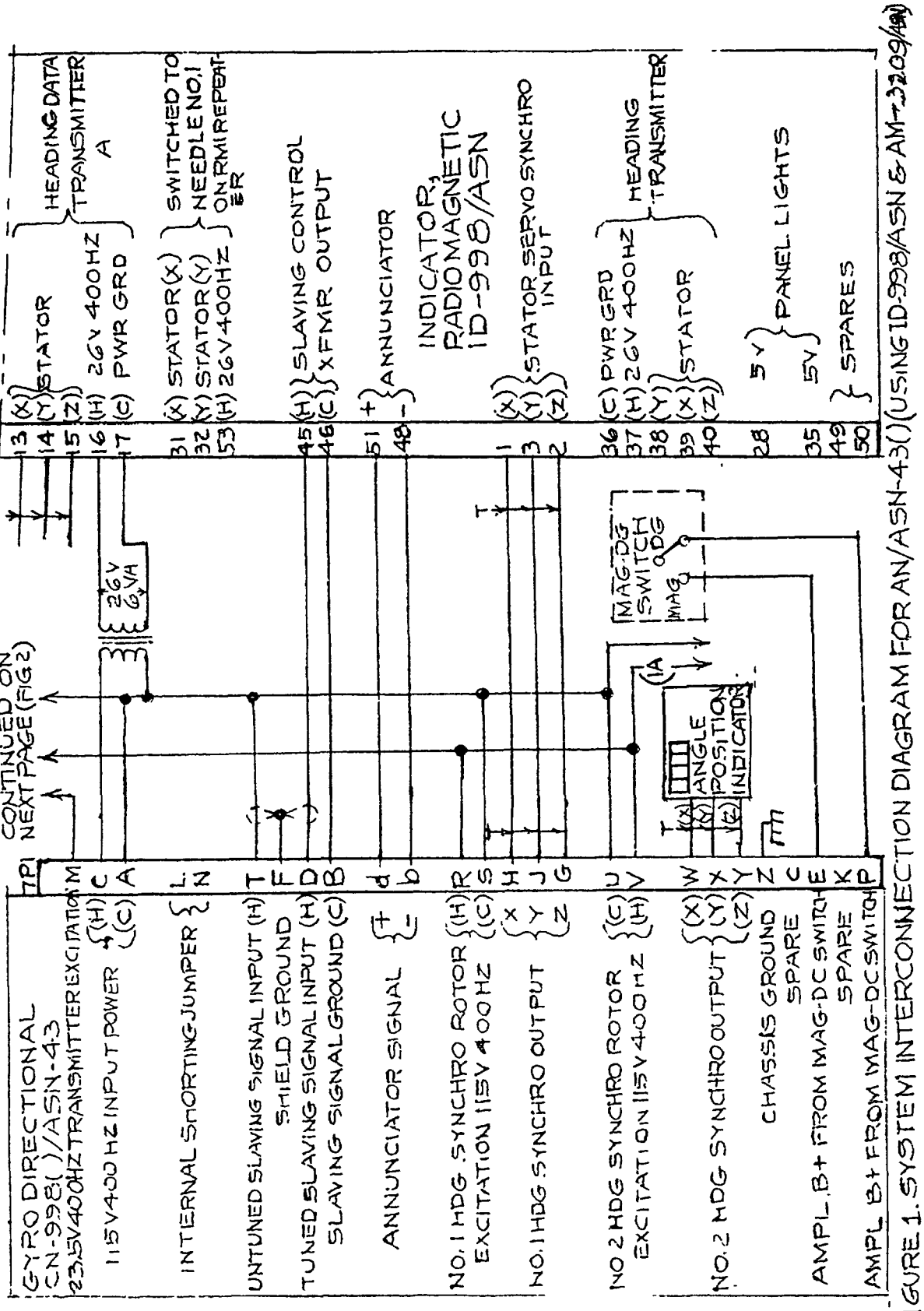


FIGURE 1. SYSTEM INTERCONNECTION DIAGRAM FOR AN/ASN-43( ) (USING ID-998/ASN & AM-3209/ASN)

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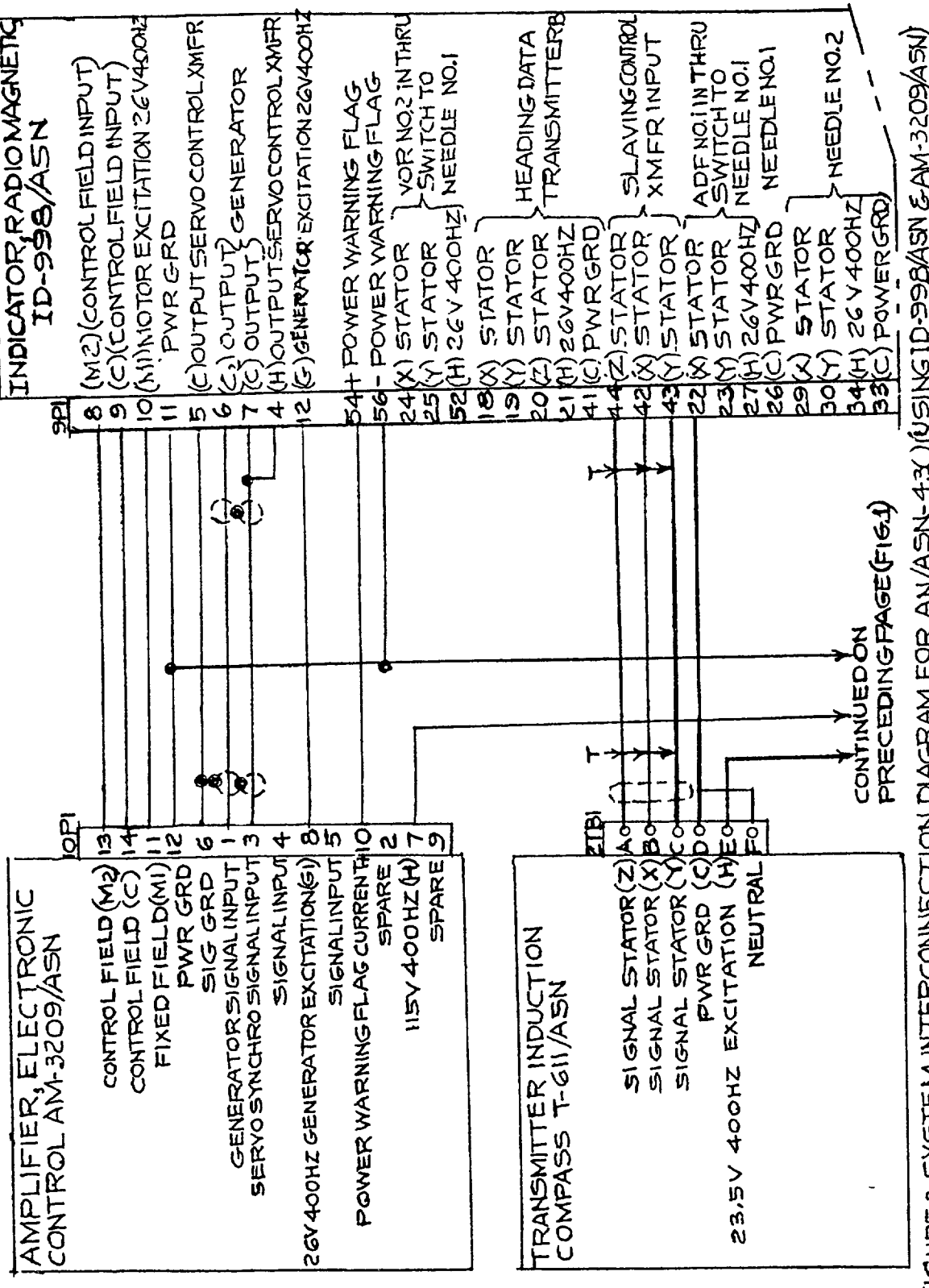


FIGURE 2. SYSTEM INTERCONNECTION DIAGRAM FOR AN/ASN-43(V) (USING ID-998/ASN & AM-3209/ASN)

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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated Fold on lines on reverse side, staple in corner, and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION MIL-G-45280(EL) GYROMAGNETIC COMPASS SET AN/ASN-43()

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING.

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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